Course introduction

Lecture 1

Course of:

Signal and imaging acquisition and modelling in environment

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General info about the course

- The course is organized in **frontal lectures** and **laboratory sessions**
  - Even during the lectures, you are often required to write some code
  - The lecture part will be recorded and made available the week after

- From the survey ~30% of you don't have any experience with ML algorithms → **we will start from the basics**

- All the exercises will be in **Python**
  - Intro to Python today
  - We will use Colab to code → shared jupyter notebooks, access to GPUs

- You can use your laptop
- Groups of **2 persons** can be formed, in that case each person is required to have its own code

Exam

- During the course we will propose some **extended exercises**
  - Most likely the time in the class won't be enough to complete them. In case you can work on them at home and ask questions if you need help.
  - You are required to **complete all the exercises**

- For the exam we ask you to:
  - **pick one extended exercise**
  - **complete it** and prepare a **written report** explaining all the details about implementation and results
  - send us the report in advance

- The exam will be **oral** and will start from the discussion of the report

- In case of groups, people belonging to the same group must pick different exercises for the written report

Content of the course and organization

The course is divided into few main chapters

1. Introduction / refresher (today)
2. 1D signals
3. From 1D to 2D: images
4. Analysis of telescope data
5. Remote sensing

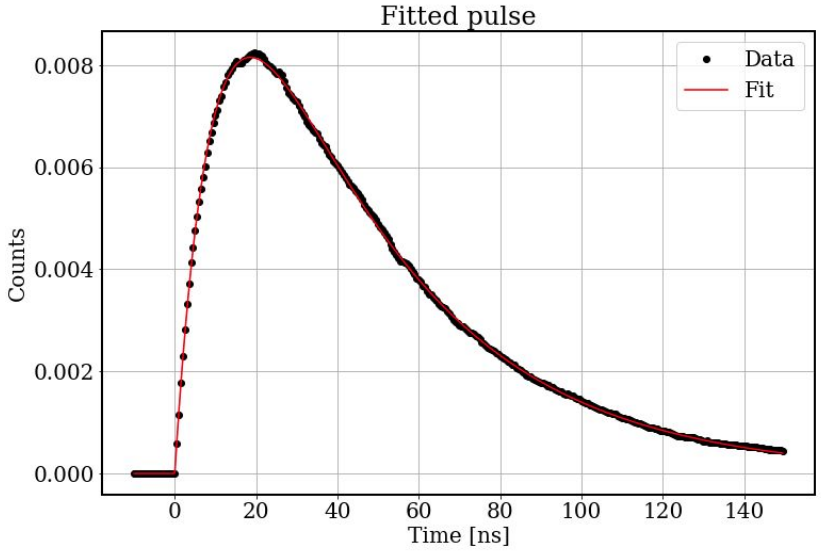
- **All topics are interconnected!**

Introduction / refresher [today]

- Intro to python
  - **Numpy, matplotlib, scipy**
- Reminder of simple notions of statistics
- How to fit a function to a dataset

1D signals

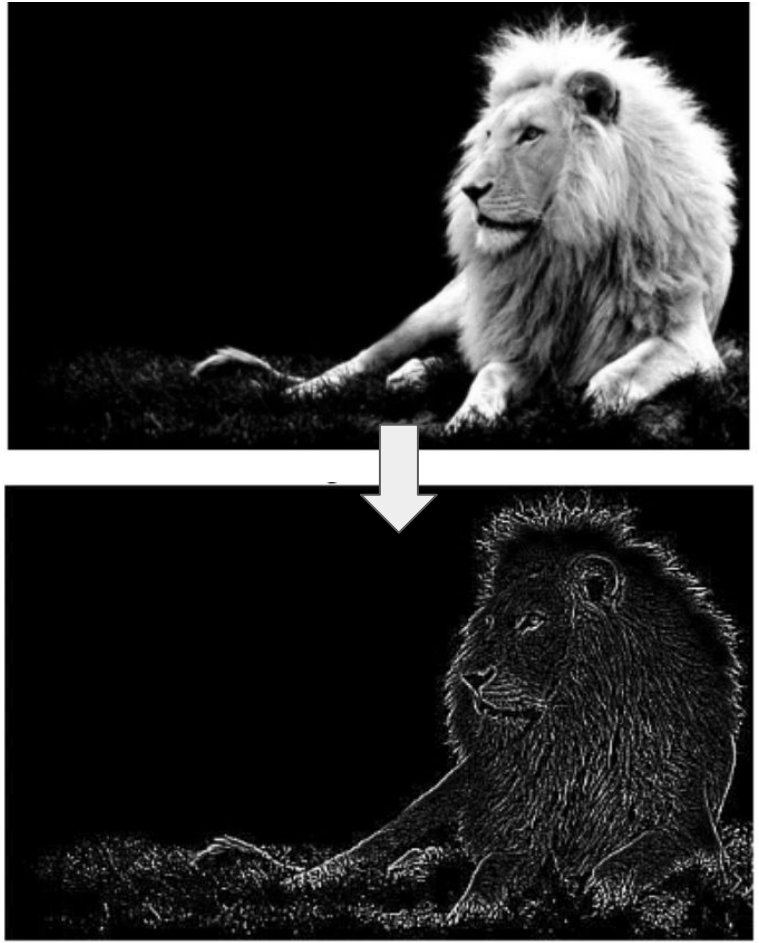
- Signals from Silicon Photomultipliers (**SiPMs**)
- Generation of a pulse library
  - The **“hit or miss” monte carlo method**
  - Simulate the impact of noise
- Write a **denoising DNN**
  - Discussion of the approach
  - Build a DNN to remove the noise from the generated samples



From 1D to 2D: images

- Intro to **OpenCV**
  - Code and test basic filters for the manipulation of the images

- Basic operations with images
  - Dynamic range
  - Picture equalization
  - Convolution of images
  - etc…

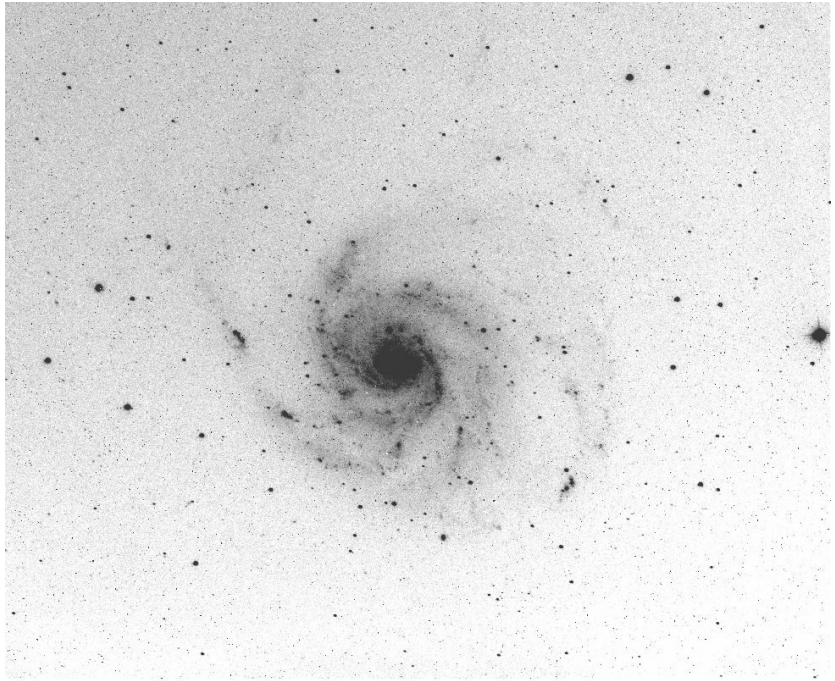


Telescope

- Experiments with images and spectra collected via the **Bicocca Optical Telescope** (on top of U9 building)

- Telescope images:
  - Characterization of the detector properties, calibrations and application to scientific images
  - Segmentation and noise properties of the images
  - **CNN exercise for galaxy classification**

- Telescope spectra:
  - Understanding optical spectra, calibrations and application to stellar spectra (Black-Body emission)
  - **CNN exercise for the extraction of stellar temperatures from spectra**



Remote sensing and segmentation

- Experiments with images collected via remote sensing

- **Drone images:**
  - Segmentation on terrain images acquired using a drone (close distance)
  - UNet architecture

- **Satellite images:**
  - Measure physical quantities on images from satellites

Intro to Python - 1

See notebooks on Colab

Intro to Python

Intro to numpy

Exercise:

- Generate 1k events gaussian distributed
- Write functions to estimate the mean and the σ of the dataset

The variance

- To measure the dispersion of the dataset, we could compute the total difference from the arithmetic mean (sum of the deviations). However this is zero by definition:

- Instead we use the **variance** defined as the **average squared sum of the deviations:**

$$s^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})^2$$

How to compute the variance

$$
\begin{aligned}
N\,s^2 &= \sum_{i=1}^{N} (x_i - \bar{x})^2 \\
&= \sum_{i=1}^{N} ({x_i}^2 + \bar{x}^2 - 2\,\bar{x}\,x_i) \\
&= \sum_{i=1}^{N} {x_i}^2 \;+\; N\,\bar{x}^2 \;-\; 2\,\bar{x} \sum_{i=1}^{N} x_i \\
&= \sum_{i=1}^{N} {x_i}^2 \;+\; N\,\bar{x}^2 \;-\; 2\,N\,\bar{x}^2 \\
&= \sum_{i=1}^{N} {x_i}^2 \;-\; N\,\bar{x}^2
\end{aligned}
$$

$$
\boxed{s^2 = \frac{1}{N} \sum_{i=1}^{N} {x_i}^2 \;-\; \bar{x}^2}
$$

Intro to Python - 2

See notebooks on Colab

Intro to Matplotlib

Estimation of parameters: what question do I want to answer?

- Let's consider two physical quantities (x, y)
- Let's assume to know the law of physics that relates the two quantities $y = f(x; \boldsymbol{a})$, but we don't know the numerical value of the parameters ($\boldsymbol{a}$)

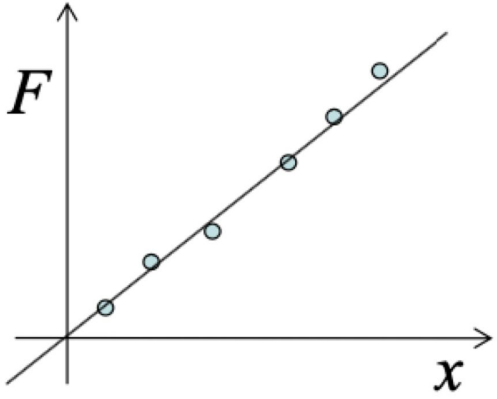
What curve best describes the data?

- We want to **fit the function to our set of measures**

Example 1

- We want to study the elastic properties of a spring
  - We fix one side of the spring and apply some tension on the other side
  - We measure how much the spring stretches as a function of the applied force
  - The Hook law predicts that the stretch is proportional to the force: **F = kx** (with $x = l - l_0$)

- Question: **what is the elastic constant of the spring?**

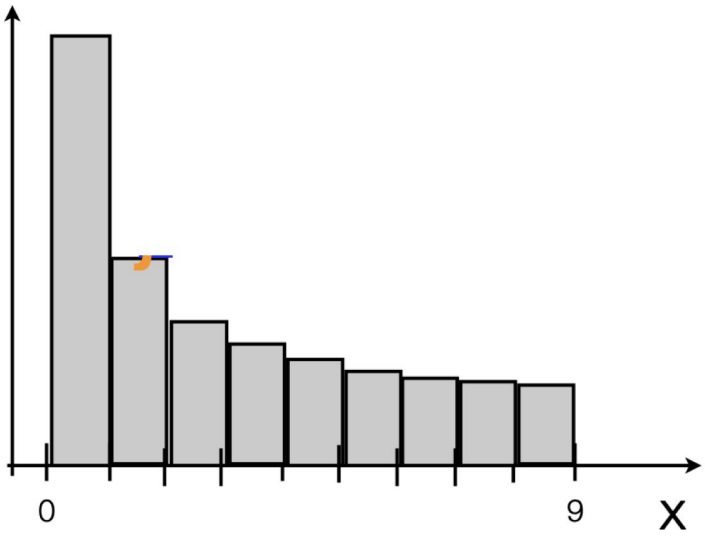


Example 2

- We want to characterize a radioactive source
  - We measure the source activity with a Geiger counter
  - The phenomenon is well described by an exponential law:

$$P(t) = \frac{1}{\tau} e^{-t/\tau}$$

- Question: **what is the value of the τ constant?**



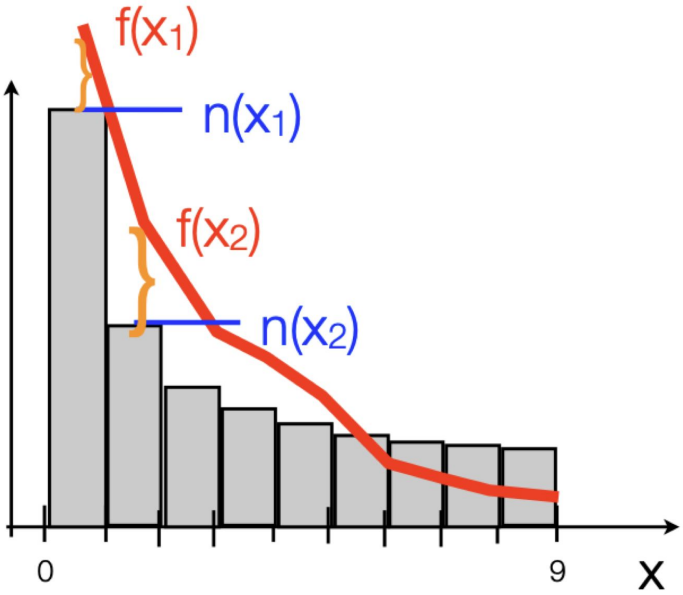
Approach: minimize the weighted distance between data and function

- When considering the data collected in an histogram we can define the distance as:

$$\chi^2 \equiv \sum_{\text{all bins}} \frac{\left(n_{\text{meas}}(x_i) - f(x_i)\right)^2}{\sigma_i^2}$$

- The error at the denominator acts as the weight

Minimization of the Chi2



Intro to Python - 3

See notebooks on Colab

Intro to SciPy

Exercise:

- Estimation of parameters

Exercise: Radioactive decay source

- Generate 1k events distributed according to a **falling exponential** with
  - Tau decay constant = 1.5 ns
  - X range = [0, 4] ns
  - N bins = 30

- **Draw the histogram** of the generated events
- **Draw the PDF** used to generate the events

- Define the fitting function
- Define the distance between the histo and the function (**Chi2**)
  - What is the error in each bin?
- **Find the best parameter(s)** that describe the data
  - Perform a parameter scan and minimize the Chi2
- Compare with the result from `optimize.curve_fit`

Reminder: the poissonian distribution

- It describes **events in a continuum**
  - Random rare events that happen at a well defined rate

- Look for the probability to observe ***r*** events in a given time interval. $\boldsymbol{\lambda}$ is the event rate.

- Example: thunderstorm
  - How many lightnings do we see in a given time interval?

$$P(r;\lambda) = e^{-\lambda}\,\frac{\lambda^r}{r!}$$

Lightning striking the Eiffel Tower, June 3, 1902, at 9:20 P.M. This is one of the earliest photographs of lightning in an urban setting In:"Thunder and Lightning", Camille Flammarion, translated by Walter Mostyn Published in 1906.

Reminder: properties of the poissonian distribution

- It is described by a **single parameter λ**

Mean	Standard deviation
λ	$\sqrt{\lambda}$